Denoising Diffusion Implicit Models (DDIMs)

Based on Song, Meng & Ermon (ICLR 2021)

Presented by Malay (as a Course Presentation for DS608)

Outline

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Background: DDPMs Recap

Setup and Notation

Notation used throughout:

- $\boldsymbol{x}_0$: clean data; $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_T$: latent variables (same dimensionality as $\boldsymbol{x}_0$)
- q : the fixed, non-parametric inference/forward process (not learned)
- p_θ : the parametric generative (reverse) process, with learnable parameters θ
- $\alpha_{1:T} \in (0, 1]^T$: a decreasing noise schedule
- $\boldsymbol{\epsilon}_\theta^{(t)} : \mathcal{X} \to \mathcal{X}$: a parametric noise-prediction network indexed by timestep t

Goal: Given samples from a data distribution $q(\boldsymbol{x}_0)$, learn a model distribution $p_\theta(\boldsymbol{x}_0)$ that approximates $q(\boldsymbol{x}_0)$ and is easy to sample from

DDPM: The Forward Process q

The forward process q is a **fixed** Markov chain that progressively adds noise:

$$q(\boldsymbol{x}_{1:T}|\boldsymbol{x}_0) = \prod_{t=1}^{T} q(\boldsymbol{x}_t|\boldsymbol{x}_{t-1}), \qquad q(\boldsymbol{x}_t|\boldsymbol{x}_{t-1}) = \mathcal{N}\Big(\sqrt{\tfrac{\alpha_t}{\alpha_{t-1}}}\,\boldsymbol{x}_{t-1},\ \Big(1 - \tfrac{\alpha_t}{\alpha_{t-1}}\Big)\,\boldsymbol{I}\Big)$$

Key property : closed-form marginals:

$$q(\boldsymbol{x}_t|\boldsymbol{x}_0) = \mathcal{N}(\sqrt{\alpha_t}\,\boldsymbol{x}_0,\ (1-\alpha_t)\boldsymbol{I})$$

equivalently,

$$\boldsymbol{x}_t = \sqrt{\alpha_t}\,\boldsymbol{x}_0 + \sqrt{1-\alpha_t}\,\boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I}).$$

When $\alpha_T \approx 0$, the endpoint $q(\boldsymbol{x}_T|\boldsymbol{x}_0) \approx \mathcal{N}(\boldsymbol{0}, \boldsymbol{I})$ for all $\boldsymbol{x}_0$.

DDPM: The Generative Process p_θ

The generative process p_θ is a **parametric** Markov chain running in reverse:

$$p_\theta(\boldsymbol{x}_{0:T}) := p_\theta(\boldsymbol{x}_T) \prod_{t=1}^{T} p_\theta^{(t)}(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t), \qquad p_\theta(\boldsymbol{x}_T) = \mathcal{N}(\mathbf{0}, \boldsymbol{I})$$

where each $p_\theta^{(t)}(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t) = \mathcal{N}\big(\mu_\theta^{(t)}(\boldsymbol{x}_t), \sigma_t^2 \boldsymbol{I}\big)$ is a learned Gaussian that approximates the intractable reverse $q(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t)$.

Training: Maximize a variational lower bound on $\log p_\theta(\boldsymbol{x}_0)$:

$$\log p_\theta(\boldsymbol{x}_0) \geq \underbrace{\mathbb{E}_{q(\boldsymbol{x}_{1:T}|\boldsymbol{x}_0)}[\log p_\theta(\boldsymbol{x}_{0:T}) - \log q(\boldsymbol{x}_{1:T}|\boldsymbol{x}_0)]}_{\text{ELBO}}$$

DDPM Training Objective: L_γ and L_1

Under Gaussian assumptions, the variational bound simplifies to minimizing:

$$L_\gamma(\theta) := \sum_{t=1}^{T} \gamma_t \, \mathbb{E}_{\boldsymbol{x}_0 \sim q(\boldsymbol{x}_0),\, \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I})} \left[\| \boldsymbol{\epsilon}_\theta^{(t)}(\underbrace{\sqrt{\alpha_t} \boldsymbol{x}_0 + \sqrt{1-\alpha_t} \boldsymbol{\epsilon}}_{\boldsymbol{x}_t}) - \boldsymbol{\epsilon} \|_2^2 \right]$$

where $\gamma = [\gamma_1, \ldots, \gamma_T]$ are positive weights determined by $\alpha_{1:T}$ and $\sigma_{1:T}$.

We write L_1 for the special case $\gamma_t = 1$ for all t:

$$\boxed{L_1(\theta) := \sum_{t=1}^{T} \mathbb{E}_{\boldsymbol{x}_0, \boldsymbol{\epsilon}} \left[\| \boldsymbol{\epsilon}_\theta^{(t)}(\boldsymbol{x}_t) - \boldsymbol{\epsilon} \|_2^2 \right]}$$

Ho et al. (2020) showed that optimizing L_1 yields the best generation quality.

Observation: L_γ depends on the **marginals** $q(\boldsymbol{x}_t | \boldsymbol{x}_0)$, *not* on the joint $q(\boldsymbol{x}_{1:T} | \boldsymbol{x}_0)$.

Motivation for DDIMs

The Sampling Bottleneck

Problem: To generate one sample from the parametric model p_θ, we must run all T reverse steps sequentially:

$$\boldsymbol{x}_T \xrightarrow{p_\theta^{(T)}} \boldsymbol{x}_{T-1} \xrightarrow{p_\theta^{(T-1)}} \cdots \xrightarrow{p_\theta^{(1)}} \boldsymbol{x}_0$$

- Typically T is in $1000s$. On a single Nvidia 2080 Ti GPU:
  - $\sim$20 hours for 50k CIFAR-10 images (32×32)
  - $\sim$1000 hours for 50k images at 256×256
- GANs: one forward pass ($<$ 1 minute for 50k CIFAR-10 images)

Goal: Sample in $S \ll T$ steps using the *same* pretrained $\boldsymbol{\epsilon}_\theta^{(t)}$

Key Insight

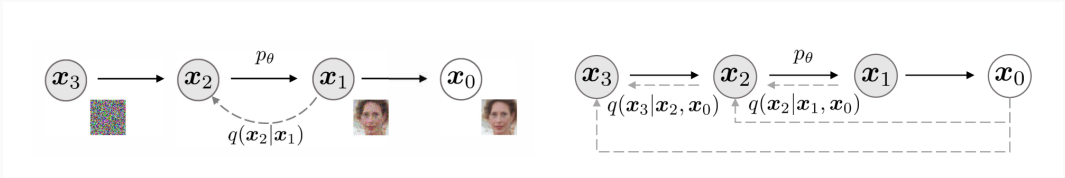
Observation (Song et al., 2021)
Since L_1 depends only on the marginals $q(\boldsymbol{x}_t|\boldsymbol{x}_0)$ and not on the joint $q(\boldsymbol{x}_{1:T}|\boldsymbol{x}_0)$, many different inference distributions share the same marginals and the same training objective.

Strategy:

1. Replace the Markovian forward process q with a **non-Markovian** one q_σ that has the same marginals.
2. Define a corresponding **new** parametric reverse process p_θ.
3. Show its variational objective equals $L_1 \implies$ **no retraining** needed.
4. Choose $\sigma = 0$ for a **deterministic** generative process.
5. Use **sub-sequences** of $\{1, \dots, T\}$ to skip steps.

Non-Markovian Forward Processes

Graphical Model: Markovian vs. Non-Markovian



Left (DDPM): each x_t depends only on x_{t-1}.

Right (DDIM): each x_t can depend on both x_{t-1} **and** x_0.

Definition For a hyperparameter $\boldsymbol{\sigma} \in \mathbb{R}^T_{\geq 0}$, define the inference process:

$$q_\sigma(\boldsymbol{x}_{1:T}|\boldsymbol{x}_0) := q_\sigma(\boldsymbol{x}_T|\boldsymbol{x}_0) \prod_{t=2}^{T} q_\sigma(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t, \boldsymbol{x}_0)$$

where $q_\sigma(\boldsymbol{x}_T|\boldsymbol{x}_0) = \mathcal{N}(\sqrt{\alpha_T}\boldsymbol{x}_0, (1-\alpha_T)\boldsymbol{I})$ and for $t > 1$:

$$q_\sigma(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t, \boldsymbol{x}_0) = \mathcal{N}\left(\sqrt{\alpha_{t-1}}\boldsymbol{x}_0 + \sqrt{1-\alpha_{t-1}-\sigma_t^2}\,\frac{\boldsymbol{x}_t - \sqrt{\alpha_t}\boldsymbol{x}_0}{\sqrt{1-\alpha_t}},\ \sigma_t^2\boldsymbol{I}\right)$$

- σ_t controls stochasticity. As $\sigma_t \to 0$: given $\boldsymbol{x}_0$ and $\boldsymbol{x}_t$, the value $\boldsymbol{x}_{t-1}$ becomes **deterministic**.
- The mean is carefully chosen to preserve the marginals (next slide).

Lemma: For $q_\sigma(\boldsymbol{x}_{1:T}|\boldsymbol{x}_0)$ defined on the previous slide, the marginals are preserved:

$$q_\sigma(\boldsymbol{x}_t|\boldsymbol{x}_0) = \mathcal{N}(\sqrt{\alpha_t}\boldsymbol{x}_0,\ (1-\alpha_t)\boldsymbol{I}) \qquad \forall\, t.$$

Proof sketch: By induction from $t = T$ to $t = 1$.

Assume $q_\sigma(\boldsymbol{x}_t|\boldsymbol{x}_0) = \mathcal{N}(\sqrt{\alpha_t}\boldsymbol{x}_0, (1-\alpha_t)\boldsymbol{I})$. Then:

$$q_\sigma(\boldsymbol{x}_{t-1}|\boldsymbol{x}_0) = \int q_\sigma(\boldsymbol{x}_t|\boldsymbol{x}_0)\ q_\sigma(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t, \boldsymbol{x}_0)\ \mathrm{d}\boldsymbol{x}_t$$

Both factors are Gaussian. The Gaussian product formula gives mean $= \sqrt{\alpha_{t-1}}\boldsymbol{x}_0$ and variance $= (1-\alpha_{t-1})\boldsymbol{I}$. □

From Baye's rule:

$$q_\sigma(\boldsymbol{x}_t \mid \boldsymbol{x}_{t-1}, \boldsymbol{x}_0) = \frac{q_\sigma(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t, \boldsymbol{x}_0)\; q_\sigma(\boldsymbol{x}_t \mid \boldsymbol{x}_0)}{q_\sigma(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_0)}$$

Since all terms are Gaussian, this simplifies to a Gaussian.

The DDIM Generative Process

Predicting x_0 from x_t

Since $\boldsymbol{x}_t = \sqrt{\alpha_t}\boldsymbol{x}_0 + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}$ and the model $\boldsymbol{\epsilon}_\theta^{(t)}(\boldsymbol{x}_t)$ predicts the noise $\boldsymbol{\epsilon}$, we can estimate the clean data:

$$\boxed{f_\theta^{(t)}(\boldsymbol{x}_t) := \frac{\boldsymbol{x}_t - \sqrt{1-\alpha_t}\,\boldsymbol{\epsilon}_\theta^{(t)}(\boldsymbol{x}_t)}{\sqrt{\alpha_t}}}$$

Here $f_\theta^{(t)}$ represents the model's best guess of $\boldsymbol{x}_0$ given $\boldsymbol{x}_t$.

The generative process p_θ then plugs $f_\theta^{(t)}(\boldsymbol{x}_t)$ in place of the unknown $\boldsymbol{x}_0$ in the reverse conditional q_σ:

$$p_\theta^{(t)}(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t) = \begin{cases} \mathcal{N}\left(f_\theta^{(1)}(\boldsymbol{x}_1),\ \sigma_1^2 \boldsymbol{I}\right) & t = 1 \\ q_\sigma\left(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t,\ f_\theta^{(t)}(\boldsymbol{x}_t)\right) & t > 1 \end{cases}$$

Sampling from the Generative Process

Sampling $\boldsymbol{x}_{t-1}$ from $\boldsymbol{x}_t$ under the reverse process p_θ (for $t > 1$):

$$\boldsymbol{x}_{t-1} = \sqrt{\alpha_{t-1}} \underbrace{\left(\frac{\boldsymbol{x}_t - \sqrt{1-\alpha_t}\, \boldsymbol{\epsilon}_\theta^{(t)}(\boldsymbol{x}_t)}{\sqrt{\alpha_t}} \right)}_{\text{“predicted } \boldsymbol{x}_0\text{”}} + \underbrace{\sqrt{1-\alpha_{t-1}-\sigma_t^2}\, \boldsymbol{\epsilon}_\theta^{(t)}(\boldsymbol{x}_t)}_{\text{“direction pointing to } \boldsymbol{x}_t\text{”}} + \underbrace{\sigma_t \boldsymbol{\epsilon}_t}_{\text{noise}}$$

where $\boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I})$ and we define $\alpha_0 := 1$.

DDPM:	$\sigma_t = \sqrt{\frac{1-\alpha_{t-1}}{1-\alpha_t}} \sqrt{1 - \frac{\alpha_t}{\alpha_{t-1}}}$	$\implies$ stochastic, Markovian reverse
DDIM:	$\sigma_t = 0 \;\; \forall t$	$\implies$ **deterministic** generation
Interpolation:	$\sigma_t(\eta) = \eta \sqrt{\frac{1-\alpha_{t-1}}{1-\alpha_t}} \sqrt{1 - \frac{\alpha_t}{\alpha_{t-1}}}$	$\eta \in (0, 1)$

Variational Objective J_σ for the Non-Markovian Process

Definition The variational objective for q_σ with reverse p_θ is:

$$J_\sigma(\theta) := \mathbb{E}_{\boldsymbol{x}_{0:T}\sim q_\sigma}[\log q_\sigma(\boldsymbol{x}_{1:T}|\boldsymbol{x}_0) - \log p_\theta(\boldsymbol{x}_{0:T})]$$

Factorizing yields:

$$J_\sigma(\theta) \equiv \mathbb{E}_{\boldsymbol{x}_{0:T}\sim q_\sigma}\left[\sum_{t=2}^{T} D_{\mathrm{KL}}\big(q_\sigma(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t, \boldsymbol{x}_0) \parallel p_\theta^{(t)}(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t)\big) - \log p_\theta^{(1)}(\boldsymbol{x}_0|\boldsymbol{x}_1)\right]$$

Each KL term compares:

- the **fixed** reverse conditional $q_\sigma(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t, \boldsymbol{x}_0)$ (uses true $\boldsymbol{x}_0$), against
- the **parametric** reverse $p_\theta^{(t)}(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t)$ (uses predicted $f_\theta^{(t)}(\boldsymbol{x}_t)$ in place of $\boldsymbol{x}_0$).

$\equiv$ denotes equality up to a constant independent of ϵ_θ

Theorem 1 (Song et al., 2021)
For all $\boldsymbol{\sigma} > 0$, there exist weights $\gamma \in \mathbb{R}^T_{>0}$ and a constant $C \in \mathbb{R}$ such that

$$J_\sigma(\theta) = L_\gamma(\theta) + C.$$

Key implication: If the parametric models $\boldsymbol{\epsilon}_\theta^{(t)}$ are *not shared* across different t, the optimal θ is the same for J_σ as for L_1, regardless of $\boldsymbol{\sigma}$.

$\implies$ **Pretrained DDPM models are already optimal for all q_σ.** No retraining.

Accelerated Sampling

Sub-Sequence Sampling

Key idea: Since L_1 only requires the marginals $q(\boldsymbol{x}_t|\boldsymbol{x}_0)$, we can define the forward process over a **sub-sequence** $\tau = [\tau_1, \ldots, \tau_S]$ of $[1, \ldots, T]$ with $S \ll T$.

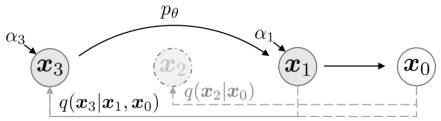


Figure 2: Graphical model for accelerated generation, where $\tau = [1, 3]$.

- The marginals $q(\boldsymbol{x}_{\tau_i}|\boldsymbol{x}_0)$ are already defined by the noise schedule.
- The pretrained parametric model $\boldsymbol{\epsilon}_\theta^{(\tau_i)}$ can be reused directly.
- Speed-up: S model evaluations instead of T.

Accelerated Inference Process $q_{\sigma,\tau}$

Given a sub-sequence $\tau = [\tau_1, \ldots, \tau_S]$, and its complement $\bar{\tau} = \{1, \ldots, T\} \setminus \tau$.

Factorized inference process:

$$q_{\sigma,\tau}(\boldsymbol{x}_{1:T}|\boldsymbol{x}_0) = q_{\sigma,\tau}(\boldsymbol{x}_{\tau_S}|\boldsymbol{x}_0) \prod_{i=1}^{S} q_{\sigma,\tau}(\boldsymbol{x}_{\tau_{i-1}}|\boldsymbol{x}_{\tau_i}, \boldsymbol{x}_0) \prod_{t\in\bar{\tau}} q_{\sigma,\tau}(\boldsymbol{x}_t|\boldsymbol{x}_0)$$

In the graphical model for $q_{\sigma,\tau}$, $\{\boldsymbol{x}_{\tau_i}\}$ form a **chain** (used for sampling) and $\{\boldsymbol{x}_t\}_{t\in\bar{\tau}}$ form a **star** with $\boldsymbol{x}_0$.

Note that the marginals are still preserved under this process.

Accelerated Generative Process and Objective

Generative process:

$$p_\theta(\boldsymbol{x}_{0:T}) = p_\theta(\boldsymbol{x}_T) \prod_{i=1}^{S} p_\theta^{(\tau_i)}(\boldsymbol{x}_{\tau_{i-1}} | \boldsymbol{x}_{\tau_i}) \prod_{t \in \bar{\tau}} p_\theta^{(t)}(\boldsymbol{x}_0 | \boldsymbol{x}_t)$$

Only chain terms are used for sampling

The Variational Objective can be written similarly as before, and still depends only on marginals $q(\boldsymbol{x}_t|\boldsymbol{x}_0)$ Hence, same ϵ_θ can be reused without retraining.

Accelerated Update Equation

For the sub-sequence τ, the parametric reverse step from $\boldsymbol{x}_{\tau_i}$ to $\boldsymbol{x}_{\tau_{i-1}}$ is:

$$\boldsymbol{x}_{\tau_{i-1}} = \sqrt{\alpha_{\tau_{i-1}}}\left(\frac{\boldsymbol{x}_{\tau_i} - \sqrt{1-\alpha_{\tau_i}}\,\boldsymbol{\epsilon}_\theta^{(\tau_i)}(\boldsymbol{x}_{\tau_i})}{\sqrt{\alpha_{\tau_i}}}\right) + \sqrt{1-\alpha_{\tau_{i-1}}-\sigma_{\tau_i}^2(\eta)}\,\boldsymbol{\epsilon}_\theta^{(\tau_i)}(\boldsymbol{x}_{\tau_i}) + \sigma_{\tau_i}(\eta)\,\boldsymbol{\epsilon}$$

where

$$\sigma_{\tau_i}(\eta) = \eta\sqrt{\frac{1-\alpha_{\tau_{i-1}}}{1-\alpha_{\tau_i}}}\sqrt{1-\frac{\alpha_{\tau_i}}{\alpha_{\tau_{i-1}}}}$$

Sub-sequence selection strategies:

- **Linear:** $\tau_i = \lfloor ci \rfloor$
- **Quadratic:** $\tau_i = \lfloor ci^2 \rfloor$

The constant c is chosen so that $\tau_S \approx T$.

Experiments

Sample Quality

Table 1: CIFAR10 and CelebA image generation measured in FID. $\eta = 1.0$ and $\hat{\sigma}$ are cases of DDPM (although Ho et al. (2020) only considered $T = 1000$ steps, and $S < T$ can be seen as simulating DDPMs trained with S steps), and $\eta = 0.0$ indicates DDIM.

		CIFAR10 (32×32)					CelebA (64×64)				
	S	10	20	50	100	1000	10	20	50	100	1000
η	0.0	**13.36**	**6.84**	**4.67**	**4.16**	4.04	**17.33**	**13.73**	**9.17**	**6.53**	3.51
	0.2	14.04	7.11	4.77	4.25	4.09	17.66	14.11	9.51	6.79	3.64
	0.5	16.66	8.35	5.25	4.46	4.29	19.86	16.06	11.01	8.09	4.28
	1.0	41.07	18.36	8.01	5.78	4.73	33.12	26.03	18.48	13.93	5.98
$\hat{\sigma}$		367.43	133.37	32.72	9.99	**3.17**	299.71	183.83	71.71	45.20	**3.26**

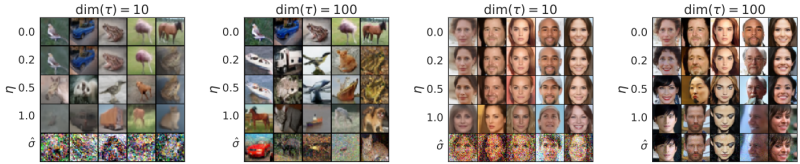


Figure 3: CIFAR10 and CelebA samples with $\dim(\tau) = 10$ and $\dim(\tau) = 100$.

Sample Quality

- DDIM ($\eta = 0$) dominates at $S \leq 100$ steps.
- 10–50$\times$ speedup with comparable quality.
- $\hat{\sigma}$: $\hat{\sigma}_{\tau_i} = \sqrt{1 - \frac{\alpha_{\tau_i}}{\alpha_{\tau_{i-1}}}}$ from Ho et al. (2020): best at $S = 1000$, but degrades for small S.
- On CelebA: FID of **100-step DDPM** $\approx$ FID of **20-step DDIM**.

Wall-Clock Time

Sampling time scales **linearly** with S.

	CIFAR-10 (50k images)	LSUN Bedroom (50k)
$S = 10$	$\sim$0.2 hours	$\sim$10 hours
$S = 100$	$\sim$2 hours	$\sim$100 hours
$S = 1000$	$\sim$20 hours	$\sim$1000 hours

(Single Nvidia 2080 Ti GPU.)

DDIM with $S = 50$–100 achieves quality close to $S = 1000$ in a fraction of the time.

Sample Consistency

DDIM ($\eta = 0$) is deterministic: the output $\boldsymbol{x}_0$ depends *only* on the initial $\boldsymbol{x}_T$.

Property: Starting from the *same* $\boldsymbol{x}_T$ with different trajectory lengths S, DDIM produces samples with **similar high-level features**.

- $S = 20$ already resembles $S = 1000$ in high-level structure.
- Trajectory length controls fine detail; $\boldsymbol{x}_T$ encodes semantics.
- DDPMs lack this: stochastic reverse produces different outputs each time.

Implication: $\boldsymbol{x}_T$ serves as a meaningful **latent code** for the generated image.

Sample Consistency



Figure 5: Samples from DDIM with the same random $\boldsymbol{x}_T$ and different number of steps.

Latent Space Interpolation

Since DDIM is deterministic, interpolating in $\boldsymbol{x}_T$-space yields semantically meaningful transitions.

- Smooth transitions in pose, expression, background.
- Analogous to GAN latent-space interpolation – without adversarial training.
- DDPMs cannot do this: same $\boldsymbol{x}_T$ leads to diverse $\boldsymbol{x}_0$.

Latent Space Interpolation

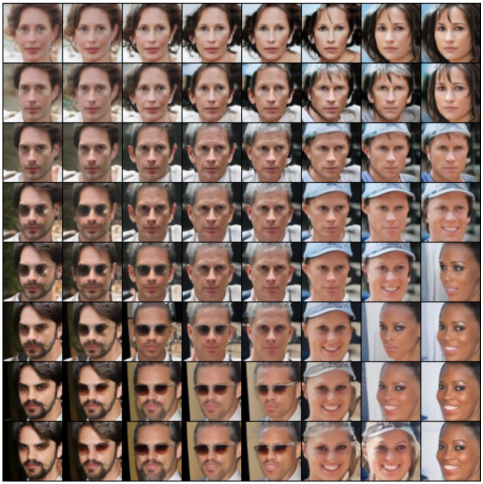


Figure 11: More interpolations from the CelebA DDIM with $\dim(\tau) = 50$.

Latent Space Interpolation



Figure 13: More interpolations from the Church DDIM with $\dim(\tau) = 50$.

Thank You!

Questions?